



Home Assignment-6 Questions

Calculus & Statistics

Instructions for solving the assignment:

- i. Every student is supposed to solve 1 sub-question from each of the main questions **A, B(i.e. total 2 questions)** based on the table given at the end of the assignment.
- ii. Get the assignment checked by your tutorial teacher on or before **14.02.25**

Q.A Solve the following

- 1) The discrete random variable x has probability distribution given by where a is a constant.

x	-1	0	1	2	3
$P(X = x)$	$\frac{1}{5}$	a	$\frac{1}{10}$	a	$\frac{1}{5}$

- (a) Find the value of a .
- (b) Find $E(X), \text{Var}(X)$.
- (c) Find $\text{Var}(Y=6-2X)$.
- (d) Calculate $P(X \geq Y)$.

2. The discrete random variable X has probability function

$$P(X = x) = \begin{cases} a(3-x) & x = 0, 1, 2 \\ b & x = 3 \end{cases}$$

Find $P(X = 2)$ and complete the table below.

X	0	1	2	3
$P(X = x)$	$3a$	$2a$		b

Given that $E(X) = 1.6$

- (a) Find the value of a and the value of b .
- (b) Find $P(0.5 < X < 3)$,
- (c) Find $E(3X - 2)$.
- (d) Find $E(X), \text{Var}(X)$.

3. When Rohit plays a game, the number of points he receives is given by the discrete random variable X with the following probability distribution.

x	0	1	2	3
$P(X = x)$	0.4	0.3	0.2	0.1

- (a) Find $E(X)$.
- (b) Find $F(1.5), F(2.3)$
- (c) $\text{Var}[(2x - 1)^2]$

(d) $P(X \geq 1)$

4. The random variable X has probability distribution given in the table below.

X	-1	0	1	2	3
$P(X = x)$	p	q	0.2	0.15	0.15

Given that $E(X) = 0.55$,

- (a) Find the value of p and the value of q
 (b) Find $\text{Var}(X)$,
 (c) Find $(2X - 4)$
 (d) $P(X \leq 1)$
5. The discrete random variable X can take only the values -1, 0 or 1. For these values the probability distribution function is defined by

$$f(x) = (x + k)^2, \quad x = -1, 0, 1$$

where k is a positive integer.

- (a) Find k .
 (b) Find the probability distribution of X .
 (c) Find $\text{Var}(X)$
 (d) $P(0 \leq X < 4)$
6. The random variable X has probability distribution

x	1	3	5	7	9
$P(X = x)$	0.2	p	0.2	q	0.15

- (a) Given that $E(X) = 4.5$, write down two equations involving p and q .
 (b) Find the value of p and the value of q ,
 (c) $P(4 < X \leq 7)$.
 (d) Find $E(X)$ and $\text{Var}(X)$
7. The random variable X has the discrete uniform distribution

$$P(X = x) = \frac{1}{5} \quad x = 1, 2, 3, 4, 5$$

- (a) Write down the value of $E(X)$ and show that $\text{Var}(X) = 2$.
 (b) Find $E(3X - 2)$,
 (c) $P(2.5 \leq X < 5)$
 (d) $\text{Var}((2X - 5)^2)$
8. The random variable X has probability distribution

x	1	2	3	4	5
$P(X = x)$	0.10	p	0.20	q	0.30

- (a) Given that $E(X) = 3.5$, write down two equations involving p and q .
 (b) Find the value of p and the value of q ,
 (c) $\text{Var}(X)$,
 (d) $\text{Var}(3 - 2X)$

9. A discrete random variable X has a probability function as shown in the table below, where a and b are constants.

x	0	1	2	3
$P(X = x)$	0.2	0.3	b	a

Given that $E(X) = 1.7$,

- find the value of a and the value of b .
- Find $P(0 < X < 1.5)$,
- $E(2X - 3)$.
- Show that $\text{Var}(X) = 1.41$

10. A discrete random variable X has the probability function shown in the table below.

x	0	1	2	3
$P(X = x)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{12}$	$\frac{1}{12}$

Find

- $P(1 < X \leq 3)$,
- $F(2.6)$,
- $E(2X - 3)$,
- $\text{Var}(X)$

11. The random variable X has the discrete uniform distribution

$$P(X = x) = \frac{1}{n}, x = 1, 2, \dots, n.$$

Given that $E(X) = 5$,

- show that $n = 9$.
- Find $P(X < 7)$,
- $\text{Var}(X)$.
- $E(3X - 4)$

12. The discrete random variable X has probability function

$$P(X = x) = \begin{cases} k(x^2 - 9) & x = 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

where k is a positive constant.

- (a) Find k
- (b) Find $E(X)$ and $\text{Var}(X)$.
- (c) Find $\text{Var}(2X - 3)$.
- (d) $P(-1 \leq X \leq 5)$

Q.B Solve the following

1. If the probability density function of a random variable is given by

$$f(x) = \begin{cases} kx^3 & 0 \leq x \leq 3 \\ 0 & \text{Otherwise} \end{cases}$$

Find the value of k and the probability that a random variable will take a value between $1/2$ and $3/2$

- (i) Mean
- (ii) Variance

2. If the probability density function of a random variable is given by

$$f(x) = \begin{cases} k(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

Find the value of k and the probability that a random variable will take a value between 0.1 and 0.2 (ii) greater than 0.5

- (i) Mean
- (ii) Variance

3. If the probability density function of a random variable is given by

$$f(x) = \begin{cases} k(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

Find the value of k and the probability that a random variable will take a value

- (i) between 0.1 and 0.2
- (ii) greater than 0.5
- (iii) Mean
- (iv) Variance

4. The daily consumption of electric power (in millions of KW-hours) is a r.v. having the pdf

$$f(x) = \begin{cases} (1/9)xe^{-x/3} & x > 0 \\ 0 & \text{Otherwise} \end{cases}$$

If the total production is 12 million KW-hours, determine the probability that there is a power cut(shortage) on any given day.

5. A continuous random variable has pdf define by

$$f(x) = \begin{cases} (1/16)(3 + x)^2 & -3 \leq x < -1 \\ (1/16)(6 - 2x^2) & -1 \leq x < 1 \\ (1/16)(3 - x)^2 & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Verify that $f(x)$ is a probability density function and also find the mean of the random variable.

6. Suppose that X is a random variable with mean μ and standard deviation σ . Find the expectation and variance of $f(x) = (x - \mu)/\sigma$.

7. A continuous random variable X has pdf defined by

$$f(x) = \begin{cases} (1/2)(x + 1) & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find mean and standard deviation of X .

8. If the probability density function of a random variable X is given by

$$f(x) = \begin{cases} kx^2 & |x| \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

Find the value of k and the probability that a random variable will take a value greater than $\frac{1}{2}$. Also find Mean and Variance of X .

9. The probability density function of a continuous random variable X is

$$f(x) = \begin{cases} x^3 & 0 < x \leq 1 \\ (2-x)^3 & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the expected value of the random variable.

10. The probability density function of a continuous random variable X is

$$f(x) = ke^{-|x|}, \quad -\infty < x < \infty$$

Find the value of k and also the mean and variance for the random variable. Also find probability between 0 and 4.

11. A continuous random variable has pdf define by

$$f(x) = \begin{cases} x^2(2x + 1.5) & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If $Y = (2/X) + 3$ find variance of Y .

12. Suppose that X is a random variable for which $E(X) = 10$ and $V(X) = 25$. Find the positive values of a and b such that $Y = aX - b$ has expectation 0 and variance 1.

Following is the question allocation according to Roll numbers

Roll No.	QA	QB
1	11	9
2	6	11
3	11	1
4	11	6
5	8	2
6	8	8
7	5	4
8	1	11
9	5	7
10	3	10
11	1	8
12	4	6
13	5	9
14	9	9
15	12	10
16	2	12
17	10	7
18	6	12
19	10	10

20	3	4
21	2	11
22	12	10
23	10	12
24	2	10
25	9	2
26	6	2
27	2	9
28	7	7
29	10	10
30	1	10
31	1	11
32	3	2
33	11	6
34	5	1
35	11	1
36	6	10
37	11	2
38	4	3
39	5	9
40	11	1
41	3	8
42	5	8
43	6	8
44	7	2
45	7	6
46	7	7
47	9	10
48	2	9
49	5	8
50	10	1
51	6	11
52	4	3
53	5	4
54	5	3
55	12	11
56	8	2
57	9	6
58	3	6
59	1	12
60	5	5
61	6	9
62	7	4
63	2	3
64	4	5
65	12	9
66	11	8
67	12	8
68	5	10

69	9	5
70	11	1
71	12	3
72	12	4
73	12	8
74	3	3
75	7	3
76	5	6
77	2	8
78	3	12